

Matteo Vissani

Research Fellow & Biomedical Engineer

✉ mvisanni@mgh.harvard.edu | 🌐 <https://matteovissani.github.io> | **in** [matteo-vissani](#) |
🌐 <https://github.com/MatteoVissani> | ☎ +1 8572108047 | 📍 Somerville, MA, USA

RESEARCH INTERESTS

My research focuses on invasive brain stimulation therapies for movement and psychiatric disorders. My aim is to uncover how targeted stimulation alters neural dynamics across distributed networks involved in movement and psychiatric disorders. My long-term goal is to develop personalized, data-driven neuromodulation strategies that move beyond empirical trial-and-error by directly linking neural activity to clinical symptoms and maladaptive behavior. This approach seeks to fundamentally transform how we understand and intervene in the brain circuits that drive severe and otherwise treatment-refractory disorders.

EDUCATION

Ph.D. in Biorobotics & Neuroengineering 2017–2021

Scuola Superiore Sant'Anna, Pisa (IT)

Dissertation: “Updating Deep Brain Stimulation: Novel quantitative methods to determine sweet spots and patient states through electrophysiological activity”

Advisors: Dr. Alberto Mazzoni, Dr. Silvestro Micera

M.Sc. in Biomedical Engineering 2014–2017

University of Bologna, Bologna (IT)

Thesis: “Multisensory features of peripersonal space representation: an analysis via neural network modelling”

Advisor: Dr. Elisa Magosso

B.Sc. in Biomedical Engineering 2011–2014

Marche Polytechnic University, Ancona (IT)

Thesis: “Ventricular fibrillation: mechanisms, time-frequency characteristics, and automatic identification”

Advisor: Dr. Laura Burattini

RESEARCH EXPERIENCE

Postdoctoral Research Fellow 06/2021 - Present

Massachusetts General Hospital, Boston (USA)

Investigating intracranial electrophysiological biomarkers of human complex behavior and psychiatric disorders.

PI: Dr. Mark Richardson

- Leading intraoperative and postoperative research with movement disorders and psychiatric patients.
- Leading data collection and analysis for multiple patient cohorts in collaboration with industrial partners.
- Developing BIDS-compliant data management pipelines for multi-stream intracranial human recordings.
- Implementing signal processing pipelines to relate neural activity to behavior and clinical symptoms.

Visiting PhD Student 10/2020 - 11/2020

Charité - Universitätsmedizin, Berlin (DE)

Investigating connectomics in Tourettian patients undergoing Deep Brain Stimulation (DBS).

PI: Dr. Andreas Horn

- Intensive training on Lead-DBS and neuroimaging tools for DBS patients.

Visiting PhD Student 02/2020 - 04/2020

University of Würzburg, Würzburg (DE)

Investigating intracranial neural oscillations during reach and grasp tasks in Parkinsonian patients.

PI: Dr. Ioannis Isaias

- Leading postoperative research in freely-moving Parkinsonian patients in lab-controlled settings.
- Implementing virtual-reality tasks to study neural activity during reach and grasp tasks.
- Implementing hardware systems to synchronize multi-modal signals in real-time.

PhD Student

10/2017 - 04/2021

Scuola Superiore Sant'Anna, Pisa (IT)

Investigating novel quantitative methods to determine sweet spots and patient states through electrophysiological activity.

PIs: Dr. Alberto Mazzoni, Dr. Silvestro Micera

- Signal processing (EMG, LFP, EEG, IMU) using Matlab, R and Python
- Image processing (f-MRI, SPECT, d-MRI) using SPM, FSL, TrackVis

Research Assistant & Neural Engineer

02/2017 - 09/2017

CHUV, Lausanne (CH)

Investigating multisensory integration deficits in amputee patients and multisensory integration illusions in healthy subjects.

PI: Dr. Andrea Serino

- Implementing an iOS app to test multisensory integration deficits in patients
- Implementing signal processing pipelines to analyze behavioral data using Matlab, R and Python
- Implementing a 3D platform for moving sounds generation for psychosocial experiments

Research Assistant & Neural Engineer

11/2016 - 03/2017

CRR SuvaCARE, Sion (CH)

Investigating multisensory integration deficits in patients with stroke using Transcranial Magnetic Stimulation (TMS) and psychosocial experiments.

PI: Dr. Michela Bassolino

- Participating in TMS sessions to record TMS-evoked motor potentials
- Implementing signal processing pipelines to analyze TMS-evoked motor potentials

Master's Degree Internship (Erasmus +)

05/2016 - 11/2016

Campus Biotech, Geneve (CH)

Internship experience to collect behavioral data for the validation of a model of multisensory integration.

PIs: Dr. Andrea Serino, Dr. Elisa Magosso, Dr. Olaf Blanke

- Leading psychosocial experiments to collect behavioral data in healthy subjects
- Learning how to implement virtual reality experiments using Oculus Rift and custom python/C# scripts (ExpyVr)

TEACHING EXPERIENCE

Teaching Assistant

2022–2024

Scuola Superiore Sant'Anna, Pisa (IT)

Professor: Dr. Alberto Mazzoni

- Neural Computing for PhD students in Biorobotics Spring 2024
- Information Theory for PhD students in Biorobotics Fall 2023
- Neural Computing for Master students in Neuroengineering Spring 2023
- Information Theory for Master students in Neuroengineering Fall 2022
- Neural Computing for Master students in Neuroengineering Spring 2022

MENTORING AND SUPERVISION EXPERIENCE

Massachusetts General Hospital, Boston (MA, USA)	2023–Present
PhD Students	
• Lucia Poma: Co-supervision with Dr. Mark Richardson and Dr. Wolf-Julian Neumann	2025 - Present
• Yanming Zhu: Co-tutoring, supervised by Dr. Mark Richardson	2023 - 2025
Dissertation: “Subthalamic nucleus participation in audition and implications for basal ganglia modulation”	
• Prince Latané Bullock: Co-tutoring, supervised by Dr. Mark Richardson	2023 - Present
Graduate Students	
• Lucia Poma: Co-tutoring, supervised by Dr. Mark Richardson and Dr. Alberto Mazzoni	Summer 2024
Thesis: “Investigation of intracranial electrophysiological dynamics of the cortico-basal ganglia circuit in Parkinsonian patients performing intraoperative speech production tasks” won the Nearlab 2025 Award (GNB – Italian National Bioengineering Group)	
Scuola Superiore Sant’Anna, Pisa (IT)	2020–2022
PhD Students	
• Ahmet Kaymak: Co-tutoring, supervised by Dr. Alberto Mazzoni	2020 - 2021
Graduate Students	
• Federico Vincenzo Mastellone: Co-tutoring, supervised by Dr. Alberto Mazzoni	2022 - 2022
Thesis: “Understanding oscillopathies in the Cortico-Basal Ganglia-Thalamocortical loop: an approach with three heterogeneous populations of Kuramoto oscillators”	
• Ahmet Kaymak: Co-tutoring, supervised by Dr. Alberto Mazzoni	2021 - 2021
Thesis: “Finding Optimal GPi Target for DBS in Dystonia Through Microelectrode Recording Analysis”	
• Elena Manferlotti: Co-tutoring, supervised by Dr. Alberto Mazzoni	2020 - 2020
Thesis: “Investigating the pathogenesis of Parkinson’s Disease in silico: correlated inputs to the striatum generate beta over-synchronization in cortico-basal ganglia network.”	
ADDITIONAL PROFESSIONAL TRAINING	
The Bok Center Certificate in Teaching , Harvard University	2025
Postdoc Leadership Course: Leadership and Management Skills Training , Harvard University	2025
Learning to Mentor and to be Mentored , Harvard University	2025
AWARDS & HONORS	
BRAIN Initiative Scholar Spotlight Award , National Institutes of Health (NIH)	2024
Michael Jenike Young Investigator Award , International OCD Foundation (IOCDF)	2023
Best Contribution Award , Workshop on AI and Smart Systems, Scuola Superiore Sant’Anna	2022
Massimo Grattarola Award , GNB – Italian National Bioengineering Group	2021
Ph.D. Awarded with Highest Honor , Scuola Superiore Sant’Anna	2021
Best 3-Minute Speech Award , Scuola Superiore Sant’Anna	2019
Ph.D. Fellowship (Top 1% of 100+ candidates) , Scuola Superiore Sant’Anna	2017–2021
Erasmus+ Scholarship , University of Bologna	2017
Merit-Based Award (1500 Euro) (Top 30 Students) , University of Bologna	2016

PEER-REVIEWED PUBLICATIONS

Robertazzi, F., **Vissani, M.**, & Falotico, E. (2025). Dynamic regulation of the serotonin-dopamine interaction within a meta-reinforcement learning framework encompassing the prefrontal cortex and basal ganglia. *International journal of neural systems*, 35(8), 2550040.

- Tafreshi, A. R., Richardson, R. M., **Vissani, M.**, Neudorfer, C., Dougherty, D. D., & Nanda, P. (2025a). 460 characterization of a candidate responsive electrophysiological biomarker in sensing-enabled deep brain stimulation of the ventral capsule/ventral striatum for obsessive-compulsive disorder. *Neurosurgery*, 71(Supplement_1), 113.
- Vissani, M.**, Bush, A., Lipski, W. J., Bullock, L., Fischer, P., Neudorfer, C., Holt, L. L., Fiez, J. A., Turner, R. S., & Mark Richardson, R. (2025). Spike-phase coupling of subthalamic neurons to posterior perisylvian cortex predicts speech sound accuracy. *Nature Communications*, 16(1), 3357.
- Hollunder, B., Ostrem, J. L., Sahin, I. A., Rajamani, N., Oxenford, S., Butenko, K., Neudorfer, C., Reinhardt, P., Zvarova, P., Polosan, M., et al. (2024). Mapping dysfunctional circuits in the frontal cortex using deep brain stimulation. *Nature Neuroscience*, 27(3), 573–586.
- Kaymak, A., Cubeddu, L., **Vissani, M.**, Taddeini, F., Caremani, L., Govoni, A., Micheli, F., Valente, S., Piattellini, F., Greco, D., et al. (2024). Low-frequency activity in dorsal subthalamic nucleus predicts impulsivity improvement following deep brain stimulation in parkinsonian patients. *medRxiv*, 2024–05.
- Ng, P. R., Bush, A., **Vissani, M.**, McIntyre, C. C., & Richardson, R. M. (2024). Biophysical principles and computational modeling of deep brain stimulation. *Neuromodulation: Technology at the Neural Interface*, 27(3), 422–439.
- Peterson, V., **Vissani, M.**, Luo, S., Rabbani, Q., Crone, N. E., Bush, A., & Richardson, R. M. (2024). A supervised data-driven spatial filter denoising method for speech artifacts in intracranial electrophysiological recordings. *Imaging Neuroscience*, 2, 1–22.
- Chua, M. M., **Vissani, M.**, Liu, D. D., Schaper, F. L., Warren, A. E., Caston, R., Dworetzky, B. A., Bubrick, E. J., Sarkis, R. A., Cosgrove, G. R., et al. (2023). Initial case series of a novel sensing deep brain stimulation device in drug-resistant epilepsy and consistent identification of alpha/beta oscillatory activity: A feasibility study. *Epilepsia*, 64(10), 2586–2603.
- Hollunder, B., Ostrem, J., Sahin, I., Rajamani, N., Oxenford, S., Butenko, K., Polosan, M., Akram, H., Vissani, M., Zhang, C., et al. (2023). Segregating the frontal cortex with deep brain stimulation. *medRxiv*, 2023–03.
- Kaymak, A., **Vissani, M.**, Lenge, M., Melani, F., Fino, E., Cappelletto, P., Tuccinardi, G., Cavallo, M. A., Giordano, F., & Mazzoni, A. (2023). Patterns of neural activity and clinical outcomes in a juvenile huntington’s disease patient undergoing deep brain stimulation of the subthalamic nucleus. *Deep Brain Stimulation*, 1, 15–19.
- Lassi, M., Dalise, S., Bandini, A., Spina, V., Azzollini, V., **Vissani, M.**, Micera, S., Mazzoni, A., & Chisari, C. (2023). Neurophysiological underpinnings of an intensive protocol for upper limb motor recovery in subacute and chronic stroke patients. *European Journal of physical and rehabilitation Medicine*, 60(1), 13.
- Neudorfer, C., Butenko, K., Oxenford, S., Rajamani, N., Achtzehn, J., Goede, L., Hollunder, B., Ríos, A. S., Hart, L., Tasserie, J., et al. (2023). Lead-dbs v3. 0: Mapping deep brain stimulation effects to local anatomy and global networks. *Neuroimage*, 268, 119862.
- Panarese, A., **Vissani, M.**, Meneghetti, N., Vannini, E., Cracchiolo, M., Micera, S., Caleo, M., Mazzoni, A., & Restani, L. (2023). Disruption of layer-specific visual processing in a model of focal neocortical epilepsy. *Cerebral Cortex*, 33(7), 4173–4187.
- Vissani, M.**, Nanda, P., Bush, A., Neudorfer, C., Dougherty, D., & Richardson, R. M. (2023). Toward closed-loop intracranial neurostimulation in obsessive-compulsive disorder. *Biological Psychiatry*, 93(11), e43–e46.
- Bassolino, M., Franza, M., Guanzioli, E., Sorrentino, G., Canzoneri, E., Colombo, M., Crema, A., Bertoni, T., Mastria, G., **Vissani, M.**, et al. (2022). Body and peripersonal space representations in chronic stroke patients with upper limb motor deficits. *Brain Communications*, 4(4), fcac179.
- Hollunder, B., Li, N., Ostrem, J., Polosan, M., Akram, H., Vissani, M., Zhang, C., Sun, B., Finke, C., Kühn, A., et al. (2022). Fv 1 segregating the prefrontal cortex by means of deep brain stimulation. *Clinical Neurophysiology*, 137, e1–e2.

- Kaymak, A., **Vissani, M.**, Rinaldo, S., Eleopra, R., Romito, L., & Mazzoni, A. (2022b). Mapping the electrophysiological structure of dystonic globus pallidus pars interna through intraoperative microelectrode recordings. *medRxiv*, 2022–11.
- Robertazzi, F., **Vissani, M.**, Schillaci, G., & Falotico, E. (2022). Brain-inspired meta-reinforcement learning cognitive control in conflictual inhibition decision-making task for artificial agents. *Neural Networks*, 154, 283–302.
- Vissani, M.**, Micheli, F., Pecchioli, G., Ramat, S., & Mazzoni, A. (2022). Impulsivity is associated with firing regularity in parkinsonian ventral subthalamic nucleus. *Annals of Clinical and Translational Neurology*, 9(4), 552–557.
- Vissani, M.**, Nanda, P., Horn, A., Bush, A., Neudorfer, C., Herrington, T., Dougherty, D., & Richardson, M. (2022). Po081/# 923 sensing-enabled ventral striatal dbs in obsessive compulsive disorder: E-poster viewing. *Neuromodulation*, 25(7), S223–S224.
- Micheli, F., **Vissani, M.**, Pecchioli, G., Terenzi, F., Ramat, S., & Mazzoni, A. (2021). Impulsivity markers in parkinsonian subthalamic single-unit activity. *Movement Disorders*, 36(6), 1435–1440.
- Vissani, M., Micheli, F., Pecchioli, G., Ramat, S., & Mazzoni, A. (2021). Markers of parkinson-related impulsive-compulsive behaviors in the activity of single units in the subthalamic nucleus: 1054. *Movement Disorders*, 36, S466.
- Vissani, M.** (2021). *Updating deep brain stimulation: Novel quantitative methods to determine sweet spots and patients state through electrophysiological activity* [Doctoral dissertation].
- Vissani, M.**, Palmisano, C., Volkmann, J., Pezzoli, G., Micera, S., Isaias, I. U., & Mazzoni, A. (2021). Impaired reach-to-grasp kinematics in parkinsonian patients relates to dopamine-dependent, subthalamic beta bursts. *npj Parkinson's Disease*, 7(1), 53.
- Marotta, G., Palmisano, C., Brandt, G., **Vissani, M.**, Pozzi, N., Canessa, A., Brumberg, J., Volkmann, J., Mazzoni, A., Frigo, C., et al. (2020). Impact of dopamine depletion on gait initiation in parkinson's disease.
- Palmisano, C., Brandt, G., **Vissani, M.**, Pozzi, N. G., Canessa, A., Brumberg, J., Marotta, G., Volkmann, J., Mazzoni, A., Pezzoli, G., et al. (2020). Gait initiation in parkinson's disease: Impact of dopamine depletion and initial stance condition. *Frontiers in bioengineering and biotechnology*, 8, 137.
- Vissani, M.**, Isaias, I. U., & Mazzoni, A. (2020). Deep brain stimulation: A review of the open neural engineering challenges. *Journal of neural engineering*, 17(5), 051002.
- Franza, M., Sorrentino, G., **Vissani, M.**, Serino, A., Blanke, O., & Bassolino, M. (2019). Hand perceptions induced by single pulse transcranial magnetic stimulation over the primary motor cortex. *Brain stimulation*, 12(3), 693–701.
- Vissani, M.**, Cordella, R., Micera, S., Eleopra, R., Romito, L. M., & Mazzoni, A. (2019). Spatio-temporal structure of single neuron subthalamic activity identifies dbs target for anesthetized tourette syndrome patients. *Journal of Neural Engineering*, 16(6), 066011.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- Molina, R., Bush, A., **Vissani, M.**, Hageman, K., Stypulkowski, P., Holt-Becker, A., Stanslaski, S., Raike, R., Herrington, T., & Richardson, R. M. (2025). P068 sensing evoked response in deep brain stimulation surgery under anesthesia. *Neuromodulation*, 28(1), S200–S201.
- Nelson, T. A., Hadar, P., Hsueh, B., Ciordia, R., Das, I., Coughlin, B., Bush, A., **Vissani, M.**, Zou, J. F., Cleary, D. R., Collins, K. M., Raslan, A. M. T., Williams, Z. M., Choi, B., Cosgrove, C. R., Dayeh, S. A., Tchoue, Y., Bourhis, A. M., Stedelin, B., ... Paulk, A. C. (2025). Cnsc-33. electrophysiologic signatures of tumor infiltration across glioma subtypes on intracranial eeg. *Neuro-Oncology*, 27(Supplement₅), v61–v61.
- Tafreshi, A. R., Richardson, R. M., **Vissani, M.**, Neudorfer, C., Dougherty, D. D., & Nanda, P. (2025b). Characterization of a candidate responsive electrophysiological biomarker in sensing-enabled deep brain stimulation of the ventral capsule/ventral striatum for obsessive-compulsive disorder. *Neurosurgery*, 71(Supplement₁), 113.

- Robertazzi, F., **Vissani, M.**, & Schillaci, G. (2023). Closed-loop brain-inspired meta-learning rules for action suppression in artificial agents. *JOURNAL OF COMPUTATIONAL NEUROSCIENCE*, 51, S37–S38.
- Kaymak, A., Vissani, M., Rinaldo, S., Eleopra, R., Romito, L., & Mazzoni, A. (2022a). Intraoperative microelectrode recordings reveal functional structure of globus pallidus pars interna in dystonic patients. *MOVEMENT DISORDERS*, 37, S277–S277.
- Manferlotti, E., **Vissani, M.**, Mazzoni, A., & Kumar, A. (2021). Correlated inputs to striatal population drive subthalamic nucleus hyper-synchronization. *2021 10th International IEEE/EMBS Conference on Neural Engineering (NER)*, 255–258.
- Robertazzi, F., Schillaci, G., **Vissani, M.**, & Falotico, E. (2021). A brain-inspired meta-reinforcement learning inhibition cognitive control for artificial agents in a conflictual decision-making task. *JOURNAL OF COMPUTATIONAL NEUROSCIENCE*, 49(SUPPL 1), S69–S69.
- Vissani, M., Cordella, R., Micera, S., Romito, L., Mazzoni, A., et al. (2019). Firing pattern of single neurons in the subthalamic nucleus of tourette syndrome patients identifies optimal deep brain stimulation target site. *Movement Disorders*, 34, S566–S567.
- Miehlbradt, J., Pierella, C., Kinany, N., Coscia, M., Pirondini, E., **Vissani, M.**, Mazzoni, A., Magnin, C., Nicolo, P., Guggisberg, A. G., et al. (2018). Evolution of cortical asymmetry with post-stroke rehabilitation: A pilot study. *International Conference on NeuroRehabilitation*, 1111–1115.
- Vissani, M.**, Serino, A., & Magosso, E. (2017). A neural network model of peripersonal space representation around different body parts. In *European medical and biological engineering conference* (pp. 217–220). Springer Singapore Singapore.

PEER-REVIEWED BOOK CHAPTERS

- Finocchiaro, M., Banfi, T., **Vissani, M.**, Mazzoni, A., Ciuti, G., et al. (2022). Healthcare (data science in), chapter 35.

PATENTS

- Inventors: A. Mazzoni, **M. Vissani** and S. Micera. “**A system for monitoring and treating motor disorders with microrecordings and targeted electrical stimulations**” Rif: 102020000026831 - Filed Nov 10th 2020

INVITED PRESENTATIONS

- 02/2024 “Distinct roles of subthalamic neuron coupling related to speech-related cortical areas”
Joint BWH-MGH Neurosurgery Translational Science Series, Virtual
- 07/2023 “Subthalamic neurons are functionally coupled to the Supramarginal gyrus”
ROH Young Investigators Meeting, Boston, MA
- 12/2022 “Speech-related spike-phase coupling of subthalamic neurons to the human cortex”
MEETs Neural Circuits Seminar Series, Boston, MA
- 10/2022 “Sensing-enabled Deep Brain Stimulation in Obsessive-Compulsive Disorder”
MGH Neuroscience Meeting, Massachusetts General Hospital, Boston, MA
- 05/2021 “High frequency subthalamic beta bursts drive reach-to-grasp kinematics in parkinsonian patients”
10th IEEE EMBS Conference on Neural Engineering, Virtual
- 09/2020 “Computational methods for model-informed neuromodulatory approaches”
Noninvasive Neuromodulation Lab Meeting, NIH, Virtual
- 03/2019 “Unveiling subthalamic nucleus functional structure to improve deep brain stimulation therapies”
BioSNS PhD Day, Scuola Normale Superiore, Pisa, IT

ORAL PRESENTATIONS OF ABSTRACTS AT CONFERENCES

- 07/2025 “Alpha Power as an OCD Biomarker in Ventral Capsule/Ventral Striatum Deep Brain Stimulation”
IOCDF Research Symposium, Chicago, IL
- 10/2024 “Characterization of a candidate responsive electrophysiological biomarker in sensing-enabled deep brain stimulation of the ventral capsule/ventral striatum for obsessive-compulsive disorder”
Congress of Neurological Surgeons Annual Meeting, Houston, TX
- 6/2024 “Electrophysiological biomarker characterization in deep brain stimulation for obsessive-compulsive disorder”
Conference Plenary Session Scholar Spotlight Talk, Bethesda, MD
- 5/2024 “Alpha power as a candidate biomarker in VC/VS DBS for obsessive-compulsive disorder”
Biennial Meeting Psychiatric Neurosurgery Symposium, Nashville, TN
- 11/2022 “Speech-related spike-phase coupling of subthalamic neurons to the human cortex”
Human Single Neuron Conference, University of California, San Francisco, CA

LEADERSHIP & SERVICE ACTIVITIES

- **Mentor**, ISSNAF 2025 - Present
Mentoring bright italian students for the ITALIAN SCIENTISTS SCHOLARS IN NORTH AMERICA FOUNDATION
- **Journal Reviewer** 2021 - Present
Nature Communications, Npj Parkinson’s Disease, Npj Digital Medicine, Brain Stimulation, Frontiers in Human Neuroscience, Communications Medicine
- **Mentor**, Lead The Future Mentorship Program 2021 - Present
Mentoring students in neuroscience and biomedical engineering
- **Conference Organizing Volunteer**, International Conference on Neurorehabilitation 2018
Assisted with conference logistics and participant coordination

SKILLS & LANGUAGES

- Programming: Python, R, MATLAB, Julia, C++, C, Bash.
- Topics: Signal processing, Information Theory, Computational modeling, Machine Learning
- Hardware: Arduino, Raspberry Pi
- Languages: Italian (native), English (fluent), French (intermediate), Spanish (intermediate).

REFERENCES

- **Dr. Darin Dougherty** (Postdoctoral Collaborator)
Director of the Division of Neurotherapeutics
Harvard Medical School - Massachusetts General Hospital
185 Cambridge St.
Boston, MA 02139
Phone: +1 617-724-6143
Email: ddougherty@mgb.org
- **Dr. Andreas Horn** (Collaborator)
Full Professor and Head of Institute for Network Stimulation
University Hospital Cologne
Kerpener Str. 62

50937, Cologne (DE)

Phone: +49 (0)221-478-30774

Email: andreas.horn@uni-koeln.de

– **Dr. Alberto Mazzoni** (PhD Supervisor)

Associate Professor and Head of the Computational Neuroengineering Lab

The Biorobotics Institute, Scuola Superiore Sant'Anna

Viale Rinaldo Piaggio 34

56025, Pontedera (IT)

Phone: +39 050-883468

Email: alberto.mazzoni@santannapisa.it

– **Dr. Silvestro Micera** (PhD collaborator)

Full Professor and Head of the Bioelectronics and Bioengineering Area

The Biorobotics Institute, Scuola Superiore Sant'Anna

Viale Rinaldo Piaggio 34

56025, Pontedera (IT)

Phone: +39 050-883454

Email: silvestro.micera@santannapisa.it

– **Dr. Michela Bassolino** (Predoctoral collaborator)

Assistant Professor and Head of the Sense Lab - Innovation & Research Center

HES-SO Valais Wallis, Institute of Health

Chemin de l'Agasse 5

1950, Sion (CH)

Phone: +41 058-606-84-04

Email: michela.bassolino@hevs.ch

– **Dr. Luigi Romito** (PhD collaborator)

Neurologist and Adjunct Professor

Carlo Besta Neurological Institute

Via Celoria 11

20133, Milan (IT)

Phone: +39 02-23942552

Email: luigi.romito@istituto-besta.it